9 March 1972

SUPERSEDING
MIL-W-3093
30 November 1949

MILITARY SPECIFICATION

CABLE, TELEPHONE INSIDE DISTRIBUTION WIRING (WD-15/U, WF-9/U, AND WT-3/U)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers 2-, 3-, and 4-conductor insulated cables, unjacketed. Each conductor consists of solid #22 AWG wire, insulated with polyvinyl chloride (PVC). The cables are for use in telephone distribution wiring systems. (See 3.2 for individual types.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-343	Wire, Electrical (Uninsulated).

MILITARY

MIL-I-3930	Insulating and Jacketing Compounds, Electrical (For Cable, Cord, and Wire).
MIL-C-12000	Cable, Cord and Wire, Electric; Packaging and Packing For.
MIL-C-45662	Calibration System Requirements.

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STANDARDS

FEDERAL

FED-STD-228 Federal Test Method Standard; Cable and Wire, Insulated; Method of Testing.

MILITARY

MIL-STD-104 Limits for Electrical Insulation Color.
MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The materials used in the fabrication of the cables shall be as specified herein.

3.2 Construction. Cables WD-15/U, WT-3/U, and WF-9/U shall be constructed with conductors of the number and size specified in table I. The conductors shall be twisted together with a right-hand lay of not less than 1-1/2 inches nor more than 3 inches.

Table I. Physical requirements.

Cable type	No. of conductors	Conductor size (AWG)	Color code
WD-15/U	2	22	Black, white
WT-3/U	3	22	Black, white, red
WF-9/U	4	22	Black, white, red, green

3.2.1 Color code. The color code for WD-15/U, WT-3/U and WF-9/U shall be as specified in table I. The specified color of each conductor shall be solid and integral throughout the insulation. The color limits for color coding shall conform to MIL-STD-104, class 1.

3.3 Conductors. The conductors shall be tinned, solid, soft or drawn-and-annealed #22 AWG copper wire conforming to QQ-W-343, type S.

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3.4 Insulation. Each conductor of cables WD-15/U, WF-9/U, and WT-3/U shall be insulated with a tightly fitting continuously extruded compound of PVC conforming to MIL-I-3930, type IP. The insulation over the conductor shall have a minimum thickness of 0.015 inches, and shall be well-centered; the minimum thickness of the insulation, measured at any cross-section along the length of the insulated conductor, shall be not less than 70 percent of the maximum thickness at that cross-section. The maximum outside diameter of each insulated conductor shall be 0.075 inches.

3.5 Electrical requirements.

3.5.1 Spark test. All insulated conductors shall withstand, without breakdown, a spark test potential of 8,000 volts (rms), when tested as specified in 4.5.2.1. A certification to the effect that the spark test has been performed shall accompany the wire when it is submitted for acceptance.

3.5.2 Dielectric withstanding voltage. The insulated conductors shall withstand, without breakdown, an applied voltage of 1000 volts (rms), when tested as specified in 4.5.2.2.

3.5.3 Insulation resistance. Immediately after the insulated conductors have been subjected to the dielectric withstanding voltage test, the insulation resistance shall be measured as specified in 4.5.2.3. The insulation resistance of the insulated conductors shall be not less than 150 megohms per 1,000 feet (single conductor).

3.5.4 Conductor resistance. Each conductor of the finished cables shall be continuous and shall have a direct current (dc) resistance of 16.1 ohms per 1,000 feet at or corrected to 20°C, when measured as specified in 4.5.2.4. If the resistance value is lower or equal to that specified when the measurement is made at a temperature greater than 20°C, no correction factor need be employed.

3.6 Physical requirements.

3.6.1 Heat shock. The insulation on the insulated conductor shall not exhibit any visible evidence of cracking when tested as specified in 4.5.3.1.

3.6.2 Heat deformation. The diameter of the insulated conductor shall not decrease more than 50 percent of the initial gage reading when tested as specified in 4.5.3.2.

3.6.3 Insulation shrinkage. (Effects of soldering.) The insulation on the insulated conductor shall not shrink back more than 5/8 inch in length from the end of the wire, when tested as specified in 4.5.3.3.

3.6.4 Flammability. The insulation on the insulated conductor shall not convey flame nor continue to burn for more than 1 minute when tested as specified in 4.5.3.4.

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3.6.5 Cold bend. The insulation on the insulated conductor shall not exhibit any evidence of cracks, flaws, or other damage when tested in accordance with 4.5.3.5.

3.7 Workmanship. Cables shall be constructed and finished in a thoroughly workmanlike manner in accordance with accepted high grade production techniques. The cables shall be a uniform and consistent product and shall be free from any defects which will adversely affect the serviceability of the product, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, skin impurities and faulty extruded surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment and facilities. Inspection equipment and facilities shall be established and maintained in accordance with MIL-C-45662.

4.2 Classification of inspections. The examinations and tests of cables are classified as follows:

(a) Materials inspection. (See 4.3.)
(b) Quality conformance inspection. (See 4.4.)
    1. Inspection of product for delivery. (See 4.4.1.)
    2. Inspection of preparation for delivery. (See 4.6.)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabricating the designated cable types (see 3.2), are in accordance with the applicable referenced specification or requirements prior to such fabrication.

Table II. Materials inspection.

Material	Requirement paragraph	Applicable specification
Conductors	3.3	QQ-W-343
Insulation	3.4	MIL-I-3930

4.4 Quality conformance inspection.

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4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of all cables of the same type (see 3.2) produced under essentially the same conditions and submitted for inspection at the one time.

4.4.1.2 Unit of product. Unless otherwise specified in the contract or order, the unit of product for purposes of sampling shall be each continuous length of cable contained on a reel, spool or coil.

4.4.1.3 Sample. The sample shall consist of that number of randomly selected units of product required by the applicable sampling plan for the presented lot.

4.4.1.4 Sample unit. A sample unit is a unit of product which forms part of the sample.

4.4.1.5 Specimen. A specimen may be the entire sample unit or any portion of the sample unit which is to be subjected to inspection.

4.4.2 Group A inspection. Group A inspection shall include the examinations and tests of table III, subgroups I and II. Statistical sampling and inspection shall be in accordance with MIL-STD-105.

4.4.2.1 Major and minor defects. Major and minor defects shall be as defined in MIL-STD-105, and as classified below. This listing is intended as a guide and should not be construed as being all-inclusive. Any sample unit which has one or more major or minor defects shall be a "defective".

Major defects:

Dimensions, conductor coating, or insulation compound not as specified.
Kinking of the bare wire within the insulated conductor.
Nicks, rips, cuts, occlusions, or similar defects in the insulation.
Insulation improperly centered over the conductor.
Incorrect insulation color code.
Insulated conductors not readily separable from each other.

Minor defects:

Insulation over the conductor not free stripping.
Improper shade of color used for coding.
Insulation surface not smooth.
Length of finished cable not in accordance with delivery instructions.

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4.4.2.2 Subgroup I. For subgroup I, the acceptable quality level (AQL) shall be as specified in table III and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup I tests may be performed in any order.

4.4.2.3 Subgroup II. For subgroup II, the AQL shall be as specified in table III, and the inspection level shall be level II in accordance with MIL-STD-105. Subgroup II tests shall be performed in the order shown in table III.

Table III. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Subgroup I				
Visual and mechanical				
Construction	3.2	4.5.1	1.0% for the subgroup	4.0% for the subgroup
Color code	3.2.1			
Conductors	3.3			
Insulation	3.4			
Workmanship	3.7			
Subgroup II				
Electrical				
Dielectric withstanding voltage	3.5.2	4.5.2.2	1.0% for the subgroup	*
Insulation resistance	3.5.3	4.5.2.3		
Conductor resistance	3.5.4	4.5.2.4		

*All electrical defects are considered major.

4.4.2.4 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Group B inspection. This inspection shall consist of the tests specified in table IV and shall be performed periodically on each different cable type (see 3.2), as indicated in 4.4.3.1. Samples shall be selected from lots that have passed group A inspection.

4.4.3.1 Sampling for inspection. One sample unit shall be selected per each 5,000 feet of cable or fraction thereof, not to exceed a total of five sample units per order. Sample units shall be selected periodically through the life of the contract to assure representative results.

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Table IV. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Heat shock	3.6.1	4.5.3.1
Heat deformation	3.6.2	4.5.3.2
Insulation shrinkage	3.6.3	4.5.3.3
Flammability	3.6.4	4.5.3.4
Cold bend	3.6.5	4.5.3.5

4.4.3.2 Noncompliance. No failures shall be allowed in group B inspection. If a sample unit fails to pass group B inspection, the supplier shall take corrective action on the material or process or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspections or the inspection that the original sample failed, at the option of the government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5 Test methods.

4.5.1 Visual and mechanical inspection. The finished cables shall be given a visual and mechanical inspection for conformance with the requirements of 3.2, 3.2.1, 3.3, 3.4, and 3.7.

4.5.2 Electrical tests.

4.5.2.1 Spark test. All insulated conductors, prior to cabling, shall be subjected to the spark test specified in FED-STD-228, method 6211, and shall meet the requirements of 3.5.1.

4.5.2.2 Dielectric withstanding voltage (see 3.5.2). The finished cables shall be tested as specified in FED-STD-228, method 6111, except for the following:

(a) The test shall be performed on finished cables only.
(b) The immersion period shall be not less than 9 hours.
(c) One terminal shall be each conductor in turn, and the other terminal shall be all the remaining conductors tied together in electrical contact with the water.

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4.5.2.3 Insulation resistance (see 3.5.3). The insulation resistance of the finished cables shall be determined as specified in FED-STD-228, method 6031, except for the following:

(a) The test shall be performed on finished cables only.
(b) The immersion period shall be not less than 9 hours.
(c) The test voltage shall be not less than 100 volts nor more than 500 volts dc.
(d) The polarity of the conductor shall be maintained negative with respect to the water. One terminal shall be each conductor in turn, and the other terminal shall be all the remaining conductors tied together in electrical contact with one another and the water.
(e) If the measurement is made at a temperature lower than 15.6°C, the supplier shall correct the measured value of insulation resistance to the resistance at 15.6°C.

4.5.2.4 Conductor resistance. The direct current (dc) resistance of each conductor in a finished cable shall be determined in accordance with FED-STD-228, method 6021, except that the immersion period shall be not less than 9 hours, and shall meet the requirements of 3.5.4.

4.5.3 Physical tests.

4.5.3.1 Heat shock. Specimens of the finished cables shall be tested in accordance with FED-STD-228, method 5011, and shall meet the requirements of 3.6.1.

4.5.3.2 Heat deformation. Specimens of the finished cables shall be tested in accordance with FED-STD-228, method 5021, except that the load on the gage shall be 200 grams, and shall meet the requirements of 3.6.2.

4.5.3.3 Insulation shrinkage. Specimens of the finished cables shall be tested in accordance with FED-STD-228, method 8231, and shall meet the requirements of 3.6.3.

4.5.3.4 Flammability. Specimens of the finished cables shall be tested in accordance with FED-STD-228, method 5221, and shall meet the requirements of 3.6.4.

4.5.3.5 Cold bend. Specimens of the finished cables shall be tested in in accordance with FED-STD-228, method 2011, including the requirements specified below, and shall meet the requirements of 3.6.5.

(a) The specimens shall be subjected to a temperature of -40°C.
(b) After the specimens and mandrels have been subjected to -40°C for 20 hours in a cold chamber, and while still in the chamber, the specimens shall be bent around a mandrel at the rate of 15 ±3 turns per minute.

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(c) The insulation on all specimens shall be examined with a magnifying glass of at least 3-diameter magnification to determine compliance with 3.6.5.

4.6 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing and marking shall be in accordance with MIL-C-12000. (See 6.2.)

6. NOTES

6.1 Intended use. The cables covered by this specification are intended for inside distribution wiring in telephone systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type designation of cable required. (See 3.2.)
(c) Length of cables to be delivered.
(d) Level of packaging and level of packing required. (See 5.1.)

6.2.1 Indirect shipments. The preservation, packaging, packing and marking specified in section 5 apply only to direct purchases by or direct shipments to the government, and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.3 Deletion of cable types W-121/U, W-122/U, W-123/U and WD-16/U. These cable types, previously referenced in MIL-W-3093, have been deleted. A suitable substitute for these types, if required, is cable type WD-15/U.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review:
Army - ME, SL
Navy - EC
Air Force - 80
NSA
DSA - IS

Users:
Army - AT

Preparing Activity:
Army - EL

Project No. 6145-0590

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098A CABLE, TELEPHONE INSIDE DISTRIBUTION WIRING (WD-15/U, WF-9/U, AND WT-3/U)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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